On a counting conjecture for binary rich words

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Abstract

Let $R_2(n)$ denote the number of binary rich words of length n. We prove that

$$\limsup_{n\to\infty} \frac{\log R_2(n)}{\sqrt{n}\log n} = \infty.$$

Consequently, $R_2(n)$ is not $O(n^{C\sqrt{n}})$ for any fixed $C > 0$, disproving the conjectured upper bound $O((n/g(n))^{\sqrt{n}})$ with $g(n) \to \infty$. The proof combines independent permutation families through an ordered concatenation construction and a binary gap encoding.

1 The result

Let $R_2(n)$ count rich words of length n over $\{0,1\}$. Guo, Shallit, and Shur [2] conjectured the bound

$$R_2(n) = O\left(\left(\frac{n}{g(n)}\right)^{\sqrt{n}}\right) \qquad \text{for some positive } g(n) \longrightarrow \infty. \tag{1}$$

This formulation also appears in [3, Section 1].

Theorem 1. *The binary rich-word counting function satisfies*

$$\limsup_{n\to\infty} \frac{\log R_2(n)}{\sqrt{n}\log n} = \infty.$$

Corollary 2. *For every $C > 0$, $R_2(n) \neq O(n^{C\sqrt{n}})$. In particular,* (1) *is false.*

Proof. An eventual bound $R_2(n) \le Kn^{C\sqrt{n}}$ would give a limsup at most C. Since $g(n) \ge 1$ eventually, (1) would imply such a bound with $C = 1$. □

The construction starts with $r!$ rich blocks of length $O(r^2)$. To multiply the choices, we place successive blocks on disjoint ordered pairs of letters and then encode the resulting word by binary gaps. The encoding charges $O(jr_j^2)$ for block j, suggesting parameters r_j proportional to $1/j$. We implement this choice dyadically: 2^t blocks of parameter $2s\,2^{k-t}$ at level t. Each level has length cost $O(s^2 4^k)$ and contributes at least $s2^k \log s$ to the logarithm of the count. Summing over $k+1$ levels gives a lower bound of order $\sqrt{k+1}$ for the normalized logarithm, with $s \to \infty$ first. All logarithms are natural.

2 Ordered concatenation

Write $\mathcal{P}(w)$ for the set of palindromic factors of w, including ε, and $D(w) = |w| + 1 - |\mathcal{P}(w)|$ for its palindromic defect. For a set S of letters, let $|w|_S$ count the positions of w carrying letters of S, and put

$$I_S(w) = \{p \in \mathcal{P}(w) : apb \text{ is a factor of } w \text{ for some } a, b \in S\},$$
$$\sigma_S(w) = \max\{|w|_S - 1, 0\} - |I_S(w)|.$$

The two endpoint letters need not coincide. We abbreviate $I_{\{b\}}$ to I_b. If w is rich and b occurs in w, then

$$|I_b(w)| = |w|_b - 1. \tag{2}$$

Each position carrying b first introduces exactly one palindrome ending in b; these are b and the words bpb with $p \in I_b(w)$ [1, proof of Proposition 2.7].

If all letters are marked and $|w| \geq 2$, then $I_{\mathrm{alph}(w)}(w)$ is the set of palindromes in w with its first and last letters deleted. Thus $\sigma_{\mathrm{alph}(w)}(w) = 0$ for rich w, also when $|w| \leq 1$.

For a word $d = a_1 \cdots a_m$ of nonnegative integers, define

$$G(d) = 0\,1^{a_1}0\,1^{a_2}0 \cdots 1^{a_m}0, \qquad G(\varepsilon) = 0.$$

For nonempty d, with $M = \max_i a_i$ and $S_h = \{a : a \geq h\}$, one has

$$D(G(d)) = D(d) + \sum_{h=1}^{M} \sigma_{S_h}(d). \tag{3}$$

To see this, the nonempty zero-ended palindromes are exactly $G(p)$, $p \in \mathcal{P}(d)$. The one-ended palindromes are $1, \ldots, 1^M$ and $1^h G(p) 1^h$ with $p \in I_{S_h}(d)$. Parsing at zeros makes these classes disjoint and exhaustive. Hence

$$|\mathcal{P}(G(d))| = 1 + |\mathcal{P}(d)| + M + \sum_{h=1}^{M} |I_{S_h}(d)|.$$

Now use $|G(d)| = 1 + m + \sum_i a_i$ and $\sum_{h=1}^{M} |d|_{S_h} = \sum_i a_i$.

Lemma 3. *Let $q \geq 1$ and let $z_1, \ldots, z_q$ be nonempty binary words ending in 1, with $z_j 1$ rich for every j. Relabel $0, 1$ in z_j as $2j-2, 2j-1$, respectively, obtaining B_j. Then $G(B_1 \cdots B_q)$ is rich.*

Proof. Put $d = B_1 \cdots B_q$. Each block is rich, and their alphabets are pairwise disjoint. No nonempty palindrome crosses a block boundary, so d is rich. We show that every term in (3) vanishes.

At an even threshold $h = 2j-2$, the marked positions form the entire rich suffix $B_j \cdots B_q$, so the marked deficit is zero. At an odd threshold $h = 2j-1$, put $U = B_j$, $b = 2j-1$, and $V = B_{j+1} \cdots B_q$. Only b is marked in U, and all letters are marked in V. If V is empty, use (2). Otherwise, writing V^- for V with its last letter deleted,

$$I_{\{b\} \cup \mathrm{alph}(V)}(UV) = I_b(Ub) \cup \mathcal{P}(V^-). \tag{4}$$

For an interior contained in U, the first letter of V can replace the auxiliary final b in Ub. For an interior contained in V, the last letter of U supplies a left endpoint. No nonempty palindrome crosses the disjoint-alphabet boundary. These observations give (4). Its two sets intersect only in ε, which belongs to both since Ub ends in bb. As Ub and V^- are rich, the union has $|U|_b + |V| - 1$ elements. The marked deficit is therefore zero. Earlier blocks contain no marked positions and do not affect these interiors. Equation (3) completes the proof. □

3 A product lower bound

Lemma 4. *For each $r \geq 1$ there are $r!$ distinct binary words z ending in 1 such that $z1$ is rich and*

$$|z| = L_r := \frac{5r^2 + 9r + 4}{2}, \qquad |z|_1 = (r+1)^2. \tag{5}$$

Proof. For a permutation $p = (p_1, \ldots, p_r)$ of $1, \ldots, r$, put $H = r + 1$ and

$$W_p = 0\,1^H \prod_{i=1}^{r} (0^i 1^{p_i} 0^i 1^H)\, 0^r.$$

The following nonempty palindromic factors occur in W_p:

$$\begin{array}{lll}
0^t, \quad 0^t 1^H 0^t & 1 \leq t \leq r, & \\
0^t 1^{p_i} 0^t & 1 \leq i \leq r, & 1 \leq t \leq i, \\
0^t 1^H 0^i 1^{p_i} 0^i 1^H 0^t & 1 \leq i \leq r, & 1 \leq t \leq \max\{1, i-1\}, \\
1^t & 1 \leq t \leq H, & \\
1^t 0^i 1^t & 1 \leq i \leq r, & 1 \leq t \leq p_i, \\
1^t 0^i 1^{p_i} 0^i 1^t & 1 \leq i \leq r, & 1 \leq t \leq H.
\end{array}$$

In the third row, the zero-run before the left 1^H has length 1 when $i = 1$ and $i - 1$ otherwise; the zero-run on the right is at least as long. This gives the stated range of t. The run patterns, $p_i < H$, and the distinct values p_i distinguish all the listed words. Their total number is $L_r = |W_p|$, so W_p is rich. Its longest zero-run has length r; appending one zero adds the new palindrome 0^{r+1}. Thus W_p0 is rich. The permutation is recoverable from the successive short one-runs, and $|W_p|_0 = (r+1)^2$. Taking z to be the bitwise complement of W_p gives the assertion. □

Proposition 5. *For $q \geq 1$ and positive integers $r_1, \ldots, r_q$, define*

$$N = 1 + \sum_{j=1}^{q} \big((2j-1)L_{r_j} + (r_j+1)^2\big).$$

Then $R_2(N) \geq \prod_{j=1}^{q} r_j!$.

Proof. Choose the words in Lemma 4 independently and apply Lemma 3. A zero of block j contributes $2j-1$ letters to the gap encoding, and a one contributes $2j$, giving the stated common length. Parsing at zeros recovers the integer word; the value $\lfloor a/2 \rfloor$ identifies the block containing each integer a. Thus every permutation is recoverable, and all outputs are distinct. □

Proof of Theorem 1. Fix $k \geq 0$. For integers $s \geq 2$, take $q = 2^{k+1} - 1$ and

$$r_j = 2s\,2^{k-t} \qquad (2^t \leq j < 2^{t+1}, \quad 0 \leq t \leq k),$$

and denote the length in Proposition 5 by $N_{s,k}$. Since $(2u)! \geq s^u$ for $u \geq s$, that proposition gives

$$\log R_2(N_{s,k}) \geq (k+1)s2^k \log s. \tag{6}$$

Also $L_r \leq 9r^2$ and $(r+1)^2 \leq 4r^2$ for $r \geq 1$. The 2^t blocks at level t have total length contribution at most

$$2^t(9 \cdot 2^{t+2} + 4)(2s\,2^{k-t})^2 \leq 160s^2 4^k.$$

The first block also gives $N_{s,k} \geq s$. Thus

$$s \leq N_{s,k} \leq 200(k+1)s^2 4^k + 1. \tag{7}$$

Letting $s \to \infty$ at fixed k in (6)–(7) gives

$$\liminf_{s\to\infty} \frac{\log R_2(N_{s,k})}{\sqrt{N_{s,k}} \log N_{s,k}} \geq \frac{\sqrt{k+1}}{2\sqrt{200}}.$$

Since $N_{s,k} \to \infty$ and k is arbitrary, the theorem follows. □

Formal verification. Theorem 1 and the negation of (1) have also been formalized in Lean 4, using the product construction and explicit integer bounds for the scaled profiles above. The formalization and paper source are available at `https://github.com/glasss13/rich-word-counting`.

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References

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